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Dapagliflozin in Patients Hospitalized for Heart Failure:

Primary Results of the DAPA ACT HF-TIMI 68 Randomized Clinical Trial and Meta-Analysis of Sodium-Glucose Cotransporter-2 Inhibitors in Patients Hospitalized for Heart Failure

David D. Berg, MD, MPH^{1,2}, Siddharth M Patel, MD, MPH^{1,2}, Paul M Haller, MD, PhD^{1,2,3}, Abby L. Cange, BS^{1,2}, Michael G Palazzolo, MS^{1,2}, Andrea Bellavia, PhD^{1,2}, Julia F. Kuder, MA^{1,2}, Akshay S. Desai, MD, MPH², Silvio E. Inzucchi, MD⁴, John J.V. McMurray, MD⁵, Eileen O'Meara, MD⁶, Subodh Verma, MD, PhD⁷, Jan Bělohávek, MD, PhD⁸, Jarosław Drożdż, MD, PhD⁹, Béla Merkely, MD, PhD¹⁰, Modele O Ogunniyi, MD, MPH¹¹, Tomáš Drasnar, MD¹², Joseph L. Izzo, MD¹³, Balazs Sarman, MD¹⁴, John E. McGinty, MD¹⁵, Krishnan Ramanathan, MB, ChB, FRACP, FRCPC¹⁶, Angel J. Mulkay, MD¹⁷, Andrzej Przybylski, MD, PhD¹⁸, Christian T. Ruff, MD, MPH^{1,2}, Michelle L. O'Donoghue, MD, MPH^{1,2}, Sabina A. Murphy, MPH^{1,2}, Marc S. Sabatine, MD, MPH^{1,2,*}, Stephen D. Wiviott, MD^{2,19,*} for the DAPA ACT HF-TIMI 68 Trial Committees and Investigators

¹TIMI Study Group, Brigham and Women's Hospital, Boston, MA, USA

²Division of Cardiovascular Medicine, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, USA

³Department of Cardiology, University Heart and Vascular Center Hamburg, University Medical Center Hamburg-Eppendorf, Hamburg, Germany

⁴Section of Endocrinology, Yale University School of Medicine, New Haven, Connecticut, USA

⁵British Heart Foundation Cardiovascular Research Centre, University of Glasgow, Glasgow, Scotland, United Kingdom

⁶Department of Cardiology, Montreal Heart Institute, Université de Montréal, Montreal, Canada

⁷Division of Cardiac Surgery, Li Ka Shing Knowledge Institute of St. Michael's Hospital, Unity Health Toronto, University of Toronto, Toronto, Canada

⁸Second Department of Internal Medicine, Cardiovascular Medicine, General University Hospital and First Faculty of Medicine, Charles University, Prague, Czech Republic; Institute for Heart Diseases, Wrocław Medical University, Wrocław, Poland

⁹Department of Cardiology, Medical University of Lodz, Lodz, Poland

¹⁰Heart and Vascular Center, Semmelweis University, Budapest, Hungary

¹¹Division of Cardiology, Department of Medicine, Emory University School of Medicine, Grady Health System, Atlanta, GA, USA

¹²Slaný Municipal Hospital, Slaný, Czech Republic

Address for Correspondence: David D. Berg, MD, MPH, Address: TIMI Study Group, Brigham and Women's Hospital, 60 Fenwood Road, Suite 7022, Boston, MA 02115, Fax: 617-734-7329, dberg1@bwh.harvard.edu.

*contributed equally

¹³Jacobs School of Medicine, University at Buffalo, Erie County Medical Center, Buffalo, NY, USA

¹⁴Department of Cardiology, Uzsoki Hospital, Budapest, Hungary

¹⁵Reid Health, Reid Physician Associates, Richmond, IN, USA

¹⁶Division of Cardiology, Department of Medicine, St Paul's Hospital and the University of British Columbia, Vancouver, BC, Canada

¹⁷Holy Name Medical Center, Teaneck, NJ, USA

¹⁸Kliniczny Szpital Wojewódzki nr 2 im. Św. Jadwigi Królowej w Rzeszowie; Kolegium Nauk Medycznych Uniwersytet Rzeszowski, Rzeszów, Poland

¹⁹Late-Stage Development, Cardiovascular, Renal and Metabolism BioPharmaceuticals Research and Development, AstraZeneca, Boston, MA, USA

Abstract

Background: Sodium-glucose cotransporter-2 inhibitors (SGLT2i) reduce the risk of cardiovascular death or worsening heart failure (HF) in outpatients with HF. Data are limited regarding initiation in patients hospitalized for HF.

Methods: We conducted a randomized, double-blind, placebo-controlled trial evaluating the efficacy and safety of in-hospital initiation of dapagliflozin (10 mg daily) in patients hospitalized for HF. The primary efficacy outcome was a composite of time to cardiovascular death or worsening HF through two months. Key safety outcomes included symptomatic hypotension and worsening kidney function. A prespecified meta-analysis was performed of randomized trials evaluating initiation of SGLT2i in hospitalized HF patients.

Results: Of 2401 patients (median age 69 [Q1-Q3, 58–77] years, 815 [33.9%] women, 448 [18.7%] Black race, 1717 [71.5%] LVEF $\leq$ 40%, 1074 [44.7%] newly diagnosed), randomized between 09/2020 and 03/2025, 1218 were assigned to dapagliflozin and 1183 to placebo. The primary outcome occurred in 133 patients (10.9%) in the dapagliflozin group and 150 (12.7%) in the placebo group (hazard ratio [HR], 0.86; 95% CI 0.68–1.08; $p=0.20$). A worsening HF event occurred in 115 (9.4%) and 122 (10.3%) patients in the dapagliflozin and placebo groups, respectively (HR, 0.91; 95% CI, 0.71–1.18). Cardiovascular death occurred in 30 (2.5%) and 37 (3.1%) patients (HR, 0.78; 95% CI, 0.48–1.27), and death from any cause occurred in 36 (3.0%) and 53 (4.5%) patients in the dapagliflozin and placebo groups, respectively (HR, 0.66; 95% CI, 0.43–1.00). The rates of symptomatic hypotension were 3.6% and 2.2%, and the rates of worsening kidney function were 5.9% and 4.7% with dapagliflozin and placebo, respectively. In a meta-analysis of patients hospitalized for HF, SGLT2i reduced the early risk of cardiovascular death or worsening HF (HR, 0.71; 95% CI, 0.54–0.93; $p=0.012$) and of all-cause death (HR, 0.57; 95% CI, 0.41–0.80; $p=0.001$).

Conclusions: In-hospital initiation of dapagliflozin did not significantly reduce the risk of cardiovascular death or worsening HF through two months in hospitalized HF patients. However, the totality of randomized clinical trial data suggests that in-hospital initiation of SGLT2i may reduce the early risk of cardiovascular death or worsening HF and of all-cause mortality.

Registration: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT04363697.

Keywords

randomized clinical trial; hospitalization; sodium-glucose cotransporter 2 inhibitors; worsening heart failure

Introduction

Hospitalization for heart failure (HF) is the leading cardiovascular reason for hospital admission globally,¹ and rates have continued to climb over the last decade.² Patients hospitalized for HF have a high risk of death and other adverse outcomes during hospital admission and in the early post-discharge period,² highlighting the need for effective treatment strategies focused on this phase of care.^{3, 4} Evidence from randomized trials suggests that initiating and optimizing disease-modifying chronic heart failure therapies, including angiotensin receptor-neprilysin inhibitors, beta-blockers, and mineralocorticoid receptor antagonists, during heart failure hospitalization and in the early post-discharge period may improve both short- and long-term outcomes.^{3, 4}

Sodium-glucose cotransporter-2 inhibitors (SGLT2i), including dapagliflozin, are indicated for the treatment of HF regardless of left ventricular ejection fraction (LVEF).^{5, 6} However, there are limited data regarding the efficacy and safety of initiating SGLT2i during HF hospitalization.^{7, 8} In the SOLOIST-WHF trial (n=1222), which enrolled patients with type 2 diabetes and a recent episode of worsening chronic HF, sotagliflozin significantly reduced total cardiovascular deaths and worsening HF events over a median follow-up of 9 months;⁹ however, only 596 patients were randomized on or before their date of discharge (76% on the date of discharge).⁸ In the EMPULSE trial (n=530), in-hospital initiation of empagliflozin in patients hospitalized for HF significantly improved a hierarchical composite outcome that included clinical events and HF-related health status as measured by the Kansas City Cardiomyopathy Questionnaire-23 item (KCCQ-23) through 90 days, though the result was driven by improvements in the latter.⁷

Data from trials of outpatients with HF suggest that the cardioprotective effects of SGLT2i emerge early, supporting prompt initiation of these therapies.^{10–12} Nevertheless, hospitalized HF patients may be more susceptible to therapy-related complications due to hemodynamic and rhythm instability, fluctuating kidney function, and concomitant addition or intensification of other HF therapies.¹³ We, therefore, designed the DAPA ACT HF-TIMI 68 (Dapagliflozin Effect on Cardiovascular Events in Acute Heart Failure – Thrombolysis in Myocardial Infarction 68) trial to test the hypothesis that in-hospital initiation of dapagliflozin, as compared with placebo, can safely and effectively decrease the early risk of cardiovascular death or worsening HF among patients hospitalized for HF. In addition, we aimed to combine data from DAPA ACT HF-TIMI 68 with other placebo-controlled trials of in-hospital initiation of SGLT2i in hospitalized HF patients to generate pooled estimates of the treatment effects of SGLT2i on early efficacy outcomes in this clinical setting.

Methods

Trial Design and Oversight

The DAPA ACT HF-TIMI 68 trial design has been reported previously.¹⁴ Briefly, DAPA ACT HF-TIMI 68 was an investigator-initiated, multinational, randomized, double-blind, placebo-controlled trial of in-hospital initiation of dapagliflozin in patients hospitalized for HF. The trial was designed by the TIMI Study Group at Brigham and Women's Hospital and Harvard Medical School (Boston, MA), in collaboration with the trial steering committee (Supplemental Material). Worldwide Clinical Trials (contract research organization) provided data management services. The safety of patients in the trial was overseen by an independent data monitoring committee. The protocol and amendments were approved by the relevant institutional review board or ethics committee at each participating site, and all participants provided written informed consent. All statistical analyses were conducted by biostatisticians at the TIMI Study Group in accordance with the statistical analysis plan (Supplemental Material). The first draft of the manuscript was written by the first author, and all authors participated in subsequent revisions. The steering committee made the decision to submit the manuscript for publication and assumes responsibility for the accuracy and completeness of the data and analyses. The manuscript adhered to the 2025 CONSORT reporting guidelines. Data will not be made publicly available; however, we encourage parties interested in collaboration and data sharing to contact the corresponding author directly for further discussion.

Participants

Eligible patients were at least 18 years of age and were currently hospitalized with a primary diagnosis of HF, including signs and symptoms of fluid overload. The trial was originally designed to enroll only patients with LVEF $\leq 40\%$; however, following publication of the EMPEROR-Preserved trial,¹⁵ which demonstrated the efficacy of the SGLT2i empagliflozin in ambulatory patients with HF and LVEF $>40\%$, the protocol was amended in November 2021 to include patients with LVEF $>40\%$. Patients were required to have elevated natriuretic peptide levels (either N-terminal pro-B-type-natriuretic peptide [NT-proBNP] or B-type-natriuretic peptide [BNP]) during the index hospitalization with varying thresholds for inclusion according to LVEF and cardiac rhythm. Patients were randomized while still hospitalized, at least 24 hours and no later than 14 days after hospital admission, and as early as possible following initial stabilization. Stabilization criteria included no intensification in the dose of intravenous diuretics during the 12 hours prior to randomization and no use of intravenous vasodilators or inotropes during the 24 hours prior to randomization. A complete list of eligibility criteria is provided in the Supplemental Material.

Procedures

Eligible patients were randomly assigned in double-blind fashion to receive dapagliflozin 10 mg once daily or matching placebo, in accordance with a sequestered, fixed randomization schedule, with the use of balanced blocks by site to ensure an approximate 1:1 allocation ratio. Randomization was stratified based on diagnosis of type 2 diabetes and history of HF prior to the index hospitalization. Open-label use of SGLT2i was prohibited during the

randomized treatment period; use of all other pharmacological HF therapies at baseline and throughout the trial was at the discretion of the treating physician. Randomized treatment was administered for 60 days, with follow-up visits scheduled at 7, 30, and 60 days after randomization. HF-related health status using the KCCQ-12 item (KCCQ-12) was assessed at randomization and the final study visit.

Outcomes

The primary efficacy outcome was a composite of time to cardiovascular death or first worsening HF event. A worsening HF event was defined as any event meeting one of the following criteria: (1) worsening signs or symptoms of HF during the index hospitalization resulting in treatment with intravenous inotropes, mechanical circulatory support, invasive ventilatory support, or heart transplantation; (2) unplanned rehospitalization for HF; or (3) urgent ambulatory HF visit resulting in treatment with intravenous diuretic therapy. A clinical events committee (CEC) led by the TIMI Study Group, whose members were blinded to randomized treatment, adjudicated all components of the primary outcome (Supplemental Material).

Secondary efficacy outcomes included a composite of time to cardiovascular death, rehospitalization for HF, or urgent HF visit; a composite of time to cardiovascular death or rehospitalization for HF; a composite of time to rehospitalization for HF or urgent HF visit; each individual component of the primary efficacy outcome; and time to death from any cause. An additional secondary outcome included a hierarchical composite of time to cardiovascular death, number of worsening HF events, time to first worsening HF event, and change from baseline in KCCQ-12 total symptom score (TSS).

Key safety outcomes included symptomatic hypotension leading to hospitalization or study drug discontinuation and worsening kidney function resulting in at least a doubling of serum creatinine, hospitalization, study drug discontinuation, dialysis, or renal death. Other prespecified safety outcomes included major hypoglycemia, diabetic ketoacidosis, adverse events leading to study drug discontinuation, and serious adverse events considered related to study drug by the investigator.

Statistical Analysis

The primary efficacy analysis was based on time from randomization to the first occurrence of any element of the primary composite outcome. We estimated that a sample size of 2400 patients would provide the trial with 80% power to detect a relative reduction of 27% in the risk of the primary outcome with dapagliflozin as compared with placebo at two months. The estimated treatment effect was based on the DAPA-HF trial of dapagliflozin in ambulatory patients with HF with reduced ejection fraction (HFrEF) as well as a subgroup analysis of DAPA-HF patients who had been hospitalized for HF in the preceding year; the relative risk reductions were 26% and 36% in the overall trial and subgroup analysis, respectively.^{10, 16} The sample size calculation assumed an event rate of 16% in the placebo group, a dropout rate of 0.5%, and a two-sided alpha level of 0.05. All efficacy analyses were conducted on an intention-to-treat basis.

The primary outcome was assessed using the log-rank test, with stratification based on diagnosis of type 2 diabetes and history of HF prior to the index hospitalization. The cumulative incidence of the primary outcome was estimated using the Kaplan-Meier method. A stratified Cox proportional-hazards model was used to estimate the treatment effect using hazard ratios (HR) and 95% confidence intervals (CI). The proportional-hazards assumption was assessed and confirmed graphically and tested with the use of Schoenfeld residuals. We assessed the consistency of the treatment effect on the primary outcome in prespecified subgroups using the Cox model. The primary outcome was also analyzed using a Bayesian survival model (Supplemental Material).

Secondary time-to-event outcomes were evaluated using stratified Cox proportional-hazards models. The secondary hierarchical composite outcome was evaluated using an unmatched win ratio approach.¹⁷ Change in KCCQ-TSS from baseline to two months in surviving patients was compared using an analysis of covariance model, adjusting for baseline value. Safety analyses were performed in patients who had undergone randomization and received at least one dose of study drug. Chi-square or Fisher's exact test was used to compare the incidence of each adverse event. Analyses were performed using SAS v9.4 and R v4.2.3.

Meta-analysis

Since DAPA ACT HF-TIMI 68 was the final planned randomized trial of in-hospital initiation of an SGLT2i, a prespecified meta-analysis was performed of randomized trials (or subgroup analyses from randomized trials) evaluating initiation of SGLT2i in patients hospitalized for HF to summarize the totality of the evidence base. Detailed methods are provided in the Supplemental Material. The main outcomes were: (1) cardiovascular death or first worsening HF event, and (2) all-cause death. Random effects models with inverse-variance weighting were used to generate estimates for the treatment effect of SGLT2i on each outcome. Between-trial heterogeneity of treatment effect was examined using Cochran's Q test.

Results

Trial Population

From September 2020 through March 2025, a total of 2401 patients underwent randomization at 210 sites in 5 countries (Figure 1), of whom 1218 were assigned to receive dapagliflozin and 1183 to receive placebo. A total of 2389 patients received at least one dose of study drug and were included in the safety analyses. A total of 262 (11.0%) patients discontinued study drug prematurely, including 139 (11.5%) in the dapagliflozin group and 123 (10.4%) in the placebo group. In total, 25 (2.1%) patients in the dapagliflozin group and 25 (2.1%) in the placebo group received at least one dose of an open-label SGLT2i during trial follow-up; however, only 9 (0.7%) and 6 (0.5%) in the dapagliflozin and placebo groups, respectively, permanently stopped study drug to start an open-label SGLT2i. Rates of withdrawal of consent (13 patients, 0.5%) and loss to follow-up with unknown vital status (4 patients, 0.2%) were low (Figure 1).

The median time from hospital admission to randomization was 3.6 days (interquartile range [IQR], 2.1–5.4 days). At the time of randomization, reported signs and symptoms of congestion were prevalent, including 39.8% with elevated jugular venous pressure, 36.4% with rales on auscultation of the lungs, and 61.8% with peripheral edema. Baseline characteristics were well balanced between the treatment groups (Table 1). The median age was 69 years (IQR, 58–77 years); 815 (33.9%) were women, 448 (18.7%) were Black, 852 (35.5%) had type 2 diabetes, and 1717 (71.5%) had a LVEF $\leq 40\%$. The index hospitalization was the first presentation of HF in 1074 (44.7%) patients. Of the remaining 1327 (55.3%) patients with worsening chronic HF, 742 (55.9%) had at least one prior HF hospitalization. At screening, the median NT-proBNP concentration (n=1582) was 4803 pg/ml (IQR, 2812–8768 pg/ml) and the median BNP concentration (n=830) was 1106 pg/ml (IQR, 717–1859 pg/ml). At randomization, the median systolic blood pressure was 119 mmHg (IQR, 108–133 mmHg), and the median eGFR was 63 ml/min/1.73m² (IQR, 48–82 ml/min/1.73m²). At randomization, 82.5% of patients were on a beta-blocker, 70.1% were on a renin-angiotensin system inhibitor, including 27.0% on an angiotensin receptor–neprilysin inhibitor (ARNI), and 48.6% were on a mineralocorticoid receptor antagonist (MRA). At discharge, 88.2% were on a beta-blocker, 75.5% were on a renin-angiotensin system inhibitor, including 30.6% on an ARNI, and 54.4% were on an MRA. The median time from randomization to hospital discharge was 2 days (IQR, 1–4 days).

Efficacy

The primary outcome occurred in 133 patients (10.9%) in the dapagliflozin group and in 150 patients (12.7%) in the placebo group (HR, 0.86; 95% CI, 0.68–1.08; p=0.20) (Table 2 and Figure 2A). There was no heterogeneity in the treatment effect of dapagliflozin on the primary outcome across prespecified subgroups (Figure 3). The results of the Bayesian analysis are presented in the Supplemental Material.

Most worsening HF events were rehospitalizations for HF, which occurred in 90 patients (7.4%) in the dapagliflozin group, as compared with 102 patients (8.6%) in the placebo group (HR, 0.85; 95% CI, 0.64–1.13) (Table 2 and Figure 2B). Cardiovascular death occurred in 30 patients (2.5%) and 37 patients (3.1%) in the dapagliflozin and placebo groups, respectively (HR, 0.78; 95% CI, 0.48–1.27) (Table 2 and Figure 2C). Death from any cause occurred in 36 patients (3.0%) and 53 patients (4.5%) in the dapagliflozin and placebo groups, respectively (HR, 0.66; 95% CI, 0.43–1.00) (Table 2 and Figure 2D).

The difference in the least-squares mean change in KCCQ-12 TSS from baseline to two months between groups was +1.3 (95% CI, –0.7 to +3.3). For the secondary hierarchical composite outcome, dapagliflozin was superior in 42.3% of paired comparisons, placebo was superior in 38.7%, and there were 19.0% ties, yielding a win ratio of 1.08 (95% CI, 0.98–1.20) (Table 2).

Safety

Symptomatic hypotension leading to hospitalization or study drug discontinuation occurred in 43 patients (3.6%) in the dapagliflozin group and 26 patients (2.2%) in the placebo group. A worsening kidney function event occurred in 71 patients (5.9%) in the dapagliflozin group

and 55 patients (4.7%) in the placebo group (Table 3). Rates of major hypoglycemia were very low and did not differ between groups. There were no cases of diabetic ketoacidosis. The rate of permanent discontinuation of study drug owing to an adverse event also did not differ between treatment groups.

Meta-analysis

Data from three trials were selected for inclusion in the meta-analysis. These studies tested dapagliflozin (DAPA ACT HF-TIMI 68),¹⁸ empagliflozin (EMPULSE),⁷ and sotagliflozin (subgroup analysis of patients initiated on study drug on or before the date of discharge in SOLOIST-WHF).⁸ Collectively, these 3 studies included 3,527 patients hospitalized for HF (Supplemental Material).^{7, 8} A total of 447 patients experienced cardiovascular death or a worsening HF event, and 153 patients died. In the meta-analysis, SGLT2i appeared to reduce the early risk of cardiovascular death or worsening HF by 29% (HR 0.71, 95% CI 0.54–0.93; $p=0.012$) and the risk of all-cause death by 43% (HR 0.57, 95% CI 0.41–0.80; $p=0.001$) (Figure 4). There was no between-trial heterogeneity of treatment effect for either outcome ($Q=3.26$ [$p=0.20$] and $Q=1.40$ [$p=0.50$], respectively).

Discussion

In this cardiovascular outcomes trial of patients hospitalized with a primary diagnosis of HF, in-hospital initiation of dapagliflozin did not significantly reduce the risk of cardiovascular death or worsening HF over two months. However, when the results of the DAPA ACT HF-TIMI 68 trial are considered in the context of other randomized trials evaluating in-hospital initiation of SGLT2i, the totality of evidence suggests a potential clinical benefit of this therapeutic class for reducing the early risks of cardiovascular death or worsening HF and of all-cause mortality when initiated in the inpatient setting.

In the SOLOIST-WHF trial, 596 patients were initiated on study drug on or before the date of discharge from their index HF hospitalization. In a *post hoc* analysis of this subgroup, sotagliflozin significantly reduced the composite of cardiovascular death or worsening HF (HR 0.54, 95% CI 0.35–0.82), and of all-cause mortality (HR 0.39, 95% CI 0.17–0.88) through 90 days.⁹ In the EMPULSE trial ($n=530$), all patients were initiated on study therapy (empagliflozin vs. placebo) while hospitalized for either newly diagnosed or worsening chronic HF, with a treatment duration of 90 days.⁷ Although EMPULSE was not powered to assess the impact of in-hospital initiation of SGLT2i therapy on clinical outcomes during and immediately following HF hospitalization, there were numerically lower rates of cardiovascular death or worsening HF (HR 0.69, 95% CI 0.45–1.08) and of all-cause mortality (HR 0.50, 95% CI 0.24–1.03) in patients randomly allocated to empagliflozin.

These studies had several important limitations, including modest sample sizes and lack of CEC adjudication of investigator-reported clinical events. Moreover, the short-term outcomes data for hospitalized HF patients from SOLOIST-WHF were based on a *post hoc* subgroup analysis, which carries a higher risk of bias compared to a full randomized trial population. In addition, there were important differences in the study populations, study designs, and follow-up durations between these two studies and the DAPA ACT

HF-TIMI 68 trial. Nevertheless, these two trials contributed to a pre-trial likelihood of clinical benefit from the strategy of in-hospital initiation of SGLT2i. In a meta-analysis using all relevant trial data, there was a significant reduction in the early risk of cardiovascular death or worsening HF with SGLT2i treatment compared to placebo, with no evidence of between-trial heterogeneity in the treatment effect and with highly consistent event rates across studies. Still, given the limitations of the prior trials, which may have contributed to overly optimistic treatment effect estimates, as well as the neutral primary result of the largest and most rigorous trial contributing to the meta-analysis, we must be cautious about the magnitude of any clinical benefit from this strategy. Indeed, it may be that the true treatment effect of SGLT2i in this clinical setting is closer to the treatment effect estimate observed in DAPA ACT HF-TIMI 68.

Several factors may have contributed to the non-significant treatment difference on the primary outcome in our trial. In contrast to EMPULSE, we designed DAPA ACT HF-TIMI 68 to have a very short two-month follow-up duration given concerns about the equipoise of a placebo control beyond that time frame in the setting of data from chronic HF trials. It is possible that the risk of worsening HF events is less modifiable and related more to patient-specific factors or models of healthcare delivery in the first month following HF hospitalization. The same pattern was observed in EMPULSE with the largest treatment effect on worsening HF events emerging between 60- and 90-days post-randomization.⁷ In addition, to consider the impact of SGLT2i on in-hospital outcomes, we included worsening HF during the index hospitalization as a component of the primary composite outcome. Although there were very few such events in our trial, there was no suggestion that dapagliflozin reduced the risk of this outcome. Finally, there were fewer than expected primary outcome events, likely due to the high proportion of newly diagnosed HF patients, which reduced the overall power to detect a significant difference in the primary outcome.

Notably, the strongest potential efficacy signal in DAPA ACT HF-TIMI 68 as well as the other two trials of in-hospital initiation of SGLT2i was the early reduction in the risk of all-cause death. The effect on mortality appears to emerge very early, suggesting that in-hospital initiation has the potential to maximize this benefit.

We observed a modestly higher risk of symptomatic hypotension leading to hospitalization or study drug discontinuation with dapagliflozin. These data suggest that in more vulnerable populations, care should be taken to minimize other potential drivers of symptomatic hypotension, such as volume depletion or rapid escalation in the dosing of neurohormonal antagonists, when initiating SGLT2i. The rate of worsening kidney function events was low but numerically higher with dapagliflozin, consistent with the known initial decline in eGFR with SGLT2i before the long-term kidney protective effects become apparent.^{19, 20} Reassuringly, the rate of major hypoglycemia was low, and there were no cases of diabetic ketoacidosis. Overall, these data are highly consistent with the known safety profile of the drug class.

Conclusions

In-hospital initiation of dapagliflozin did not significantly reduce the risk of cardiovascular death or worsening HF over two months in patients admitted for HF in the DAPA ACT HF-TIMI 68 trial. However, the totality of randomized clinical trial data suggests that in-hospital initiation of SGLT2i may reduce the high risk of clinical events, including all-cause mortality, in the early post-discharge period.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Sources of Funding

AstraZeneca provided study drug and funding for the trial.

Disclosures

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Non-standard Abbreviations and Acronyms

ARNI	angiotensin receptor–neprilysin inhibitor
BNP	B-type-natriuretic peptide
CEC	clinical events committee
DAPA ACT HF-TIMI 68	Dapagliflozin Effect on Cardiovascular Events in Acute Heart Failure – Thrombolysis in Myocardial Infarction 68
HF	heart failure
HFrEF	heart failure with reduced ejection fraction
KCCQ	Kansas City Cardiomyopathy Questionnaire
LVEF	left ventricular ejection fraction
MRA	mineralocorticoid receptor antagonist
NT-proBNP	N-terminal pro-B-type-natriuretic peptide
SGLT2i	sodium-glucose cotransporter-2 inhibitors
TIMI	Thrombolysis in Myocardial Infarction
TSS	total symptom score

References

1. Martin SS, Aday AW, Allen NB, Almarzooq ZI, Anderson CAM, Arora P, Avery CL, Baker-Smith CM, Bansal N, Beaton AZ, Commodore-Mensah Y, Currie ME, Elkind MSV, Fan W, Generoso G, Gibbs BB, Heard DG, Hiremath S, Johansen MC, Kazi DS, Ko D, Leppert MH, Magnani JW, Michos ED, Mussolino ME, Parikh NI, Perman SM, Rezk-Hanna M, Roth GA, Shah NS, Springer

- MV, St-Onge MP, Thacker EL, Urbut SM, Van Spall HGC, Voeks JH, Whelton SP, Wong ND, Wong SS, Yaffe K, Palaniappan LP, American Heart Association Council on E, Prevention Statistics C and Stroke Statistics C. 2025 Heart Disease and Stroke Statistics: A Report of US and Global Data From the American Heart Association. *Circulation*. 2025;151:e41–e660. [PubMed: 39866113]
2. Bozkurt B, Ahmad T, Alexander K, Baker WL, Bosak K, Brethett K, Carter S, Drazner MH, Dunlay SM, Fonarow GC, Greene SJ, Heidenreich P, Ho JE, Hsieh E, Ibrahim NE, Jones LM, Khan SS, Khazanie P, Koelling T, Lee CS, Morris AA, Page RL 2nd, Pandey A, Piano MR, Sandhu AT, Stehlik J, Stevenson LW, Teerlink J, Vest AR, Yancy C, Ziaeian Band Writing Committee M. HF STATS 2024: Heart Failure Epidemiology and Outcomes Statistics An Updated 2024 Report from the Heart Failure Society of America. *J Card Fail*. 2025;31:66–116. [PubMed: 39322534]
 3. Mebazaa A, Davison B, Chioncel O, Cohen-Solal A, Diaz R, Filippatos G, Metra M, Ponikowski P, Sliwa K, Voors AA, Edwards C, Novosadova M, Takagi K, Damasceno A, Saidu H, Gayat E, Pang PS, Celutkienė J and Cotter G. Safety, tolerability and efficacy of up-titration of guideline-directed medical therapies for acute heart failure (STRONG-HF): a multinational, open-label, randomised, trial. *Lancet*. 2022;400:1938–1952. [PubMed: 36356631]
 4. Morrow DA, Velazquez EJ, DeVore AD, Desai AS, Duffy CI, Ambrosy AP, Gurmu Y, McCague K, Rocha R and Braunwald E. Clinical Outcomes in Patients With Acute Decompensated Heart Failure Randomly Assigned to Sacubitril/Valsartan or Enalapril in the PIONEER-HF Trial. *Circulation*. 2019;139:2285–2288. [PubMed: 30955360]
 5. Heidenreich PA, Bozkurt B, Aguilar D, Allen LA, Byun JJ, Colvin MM, Deswal A, Drazner MH, Dunlay SM, Evers LR, Fang JC, Fedson SE, Fonarow GC, Hayek SS, Hernandez AF, Khazanie P, Kittleson MM, Lee CS, Link MS, Milano CA, Nwacheta LC, Sandhu AT, Stevenson LW, Vardeny O, Vest AR, Yancy CW and Members AAJC. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145:e895–e1032. [PubMed: 35363499]
 6. McDonagh TA, Metra M, Adamo M, Gardner RS, Baumbach A, Böhm M, Burri H, Butler J, Celutkienė J, Chioncel O, Cleland JGF, Crespo-Leiro MG, Farmakis D, Gilard M, Heymans S, Hoes AW, Jaarsma T, Jankowska EA, Lainscak M, Lam CSP, Lyon AR, McMurray JJV, Mebazaa A, Mindham R, Muneretto C, Francesco Piepoli M, Price S, Rosano GMC, Ruschitzka F, Skibellund AK and Group ESCSD. 2023 Focused Update of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2023;44:3627–3639. [PubMed: 37622666]
 7. Voors AA, Angermann CE, Teerlink JR, Collins SP, Kosiborod M, Biegus J, Ferreira JP, Nassif ME, Psotka MA, Tromp J, Borleffs CJW, Ma C, Comin-Colet J, Fu M, Janssens SP, Kiss RG, Mentz RJ, Sakata Y, Schirmer H, Schou M, Schulze PC, Spinarova L, Volterrani M, Wranicz JK, Zeymer U, Zieroth S, Brueckmann M, Blatchford JP, Salsali A and Ponikowski P. The SGLT2 inhibitor empagliflozin in patients hospitalized for acute heart failure: a multinational randomized trial. *Nat Med*. 2022;28:568–574. [PubMed: 35228754]
 8. Pitt B, Bhatt DL, Szarek M, Cannon CP, Leiter LA, McGuire DK, Lewis JB, Riddle MC, Voors AA, Metra M, Lund LH, Komajda M, Testani JM, Wilcox CS, Ponikowski P, Lopes RD, Ezekowitz JA, Sun F, Davies MJ, Verma S, Kosiborod MN, Steg PG and Investigators S-W. Effect of Sotagliflozin on Early Mortality and Heart Failure-Related Events: A Post Hoc Analysis of SOLOIST-WHF. *JACC Heart Fail*. 2023;11:879–889. [PubMed: 37558385]
 9. Bhatt DL, Szarek M, Steg PG, Cannon CP, Leiter LA, McGuire DK, Lewis JB, Riddle MC, Voors AA, Metra M, Lund LH, Komajda M, Testani JM, Wilcox CS, Ponikowski P, Lopes RD, Verma S, Lapuerta P, Pitt B and Investigators S-WT. Sotagliflozin in Patients with Diabetes and Recent Worsening Heart Failure. *N Engl J Med*. 2021;384:117–128. [PubMed: 33200892]
 10. Berg DD, Jhund PS, Docherty KF, Murphy SA, Verma S, Inzucchi SE, Kober L, Kosiborod MN, Langkilde AM, Martinez FA, Bengtsson O, Ponikowski P, Sjostrand M, Solomon SD, McMurray JJV and Sabatine MS. Time to Clinical Benefit of Dapagliflozin and Significance of Prior Heart Failure Hospitalization in Patients With Heart Failure With Reduced Ejection Fraction. *JAMA Cardiol*. 2021;6:499–507. [PubMed: 33595593]
 11. Vaduganathan M, Claggett BL, Jhund P, de Boer RA, Hernandez AF, Inzucchi SE, Kosiborod MN, Lam CSP, Martinez F, Shah SJ, Desai AS, Hegde SM, Lindholm D, Petersson M, Langkilde AM, McMurray JJV and Solomon SD. Time to Clinical Benefit of Dapagliflozin in Patients With Heart Failure With Mildly Reduced or Preserved Ejection Fraction: A Prespecified Secondary Analysis

of the DELIVER Randomized Clinical Trial. *JAMA Cardiol.* 2022;7:1259–1263. [PubMed: 36190011]

12. Packer M, Anker SD, Butler J, Filippatos G, Ferreira JP, Pocock SJ, Carson P, Anand I, Doehner W, Haass M, Komajda M, Miller A, Pehrson S, Teerlink JR, Brueckmann M, Jamal W, Zeller C, Schnaidt S and Zannad F. Effect of Empagliflozin on the Clinical Stability of Patients With Heart Failure and a Reduced Ejection Fraction: The EMPEROR-Reduced Trial. *Circulation.* 2021;143:326–336. [PubMed: 33081531]
13. Gheorghiadu M, Vaduganathan M, Fonarow GC and Bonow RO. Rehospitalization for heart failure: problems and perspectives. *J Am Coll Cardiol.* 2013;61:391–403. [PubMed: 23219302]
14. Berg DD, Patel SM, Haller PM, Belohlavek J, Desai AS, Drozd J, Inzucchi SE, McMurray JJV, Merkely B, O'Meara E, Verma S, Cange AL, Murphy SA, Sabatine MS and Wiviott SD. Rationale and Design of the Dapagliflozin Effect on Cardiovascular Events in Acute Heart Failure (DAPA ACT HF)-TIMI 68 Trial. *JACC Heart Fail.* 2025;13:829–839. [PubMed: 40335233]
15. Anker SD, Butler J, Filippatos G, Ferreira JP, Bocchi E, Bohm M, Brunner-La Rocca HP, Choi DJ, Chopra V, Chuquiere-Valenzuela E, Giannetti N, Gomez-Mesa JE, Janssens S, Januzzi JL, Gonzalez-Juanatey JR, Merkely B, Nicholls SJ, Perrone SV, Pina IL, Ponikowski P, Senni M, Sim D, Spinar J, Squire I, Taddei S, Tsutsui H, Verma S, Vinereanu D, Zhang J, Carson P, Lam CSP, Marx N, Zeller C, Sattar N, Jamal W, Schnaidt S, Schnee JM, Brueckmann M, Pocock SJ, Zannad F, Packer M and Investigators EM-PT. Empagliflozin in Heart Failure with a Preserved Ejection Fraction. *N Engl J Med.* 2021;385:1451–1461. [PubMed: 34449189]
16. McMurray JJV, Solomon SD, Inzucchi SE, Kober L, Kosiborod MN, Martinez FA, Ponikowski P, Sabatine MS, Anand IS, Belohlavek J, Bohm M, Chiang CE, Chopra VK, de Boer RA, Desai AS, Diez M, Drozd J, Dukat A, Ge J, Howlett JG, Katova T, Kitakaze M, Ljungman CEA, Merkely B, Nicolau JC, O'Meara E, Petrie MC, Vinh PN, Schou M, Tereshchenko S, Verma S, Held C, DeMets DL, Docherty KF, Jhund PS, Bengtsson O, Sjostrand M, Langkilde AM, Committees D-HT and Investigators. Dapagliflozin in Patients with Heart Failure and Reduced Ejection Fraction. *N Engl J Med.* 2019;381:1995–2008. [PubMed: 31535829]
17. Pocock SJ, Ariti CA, Collier TJ and Wang D. The win ratio: a new approach to the analysis of composite endpoints in clinical trials based on clinical priorities. *Eur Heart J.* 2012;33:176–82. [PubMed: 21900289]
18. Berg DD, Patel SM, Haller PM, Belohlávek J, Desai AS, Drozd J, Inzucchi SE, McMurray JJV, Merkely B, O'Meara E, Verma S, Cange AL, Murphy SA, Sabatine MS and Wiviott SD. Rationale and Design of the Dapagliflozin Effect on Cardiovascular Events in Acute Heart Failure (DAPA ACT HF)-TIMI 68 Trial. *JACC: Heart Failure.* 2025;13:829–839. [PubMed: 40335233]
19. Heerspink HJL, Stefansson BV, Correa-Rotter R, Chertow GM, Greene T, Hou FF, Mann JFE, McMurray JJV, Lindberg M, Rossing P, Sjoström CD, Toto RD, Langkilde AM, Wheeler DC, Committees D-CT and Investigators. Dapagliflozin in Patients with Chronic Kidney Disease. *N Engl J Med.* 2020;383:1436–1446. [PubMed: 32970396]
20. The E-KCG, Herrington WG, Staplin N, Wanner C, Green JB, Hauske SJ, Emberson JR, Preiss D, Judge P, Mayne KJ, Ng SYA, Sammons E, Zhu D, Hill M, Stevens W, Wallendszus K, Brenner S, Cheung AK, Liu ZH, Li J, Hooi LS, Liu W, Kadowaki T, Nangaku M, Levin A, Cherney D, Maggioni AP, Pontremoli R, Deo R, Goto S, Rossello X, Tuttle KR, Steubl D, Petrini M, Massey D, Eilbracht J, Brueckmann M, Landray MJ, Baigent C and Haynes R. Empagliflozin in Patients with Chronic Kidney Disease. *N Engl J Med.* 2023;388:117–127. [PubMed: 36331190]
21. Roychoudhury S and Neuenschwander B. Bayesian leveraging of historical control data for a clinical trial with time-to-event endpoint. *Stat Med.* 2020;39:984–995. [PubMed: 31985077]

Clinical Perspective

What is new?

- DAPA ACT HF-TIMI 68 was a randomized, double-blind, placebo-controlled trial evaluating the efficacy and safety of in-hospital initiation of dapagliflozin in patients hospitalized for heart failure (HF).
- In this trial, in-hospital initiation of dapagliflozin did not significantly reduce the risk of cardiovascular death or worsening HF through 2 months (hazard ratio [HR] 0.86, 95% confidence interval [CI] 0.68–1.08).
- In a prespecified meta-analysis of three trials evaluating in-hospital initiation of sodium-glucose cotransporter-2 inhibitors (SGLT2i) in patients hospitalized for HF, SGLT2i appeared to reduce the early risk of cardiovascular death or worsening HF (HR 0.71; 95% CI 0.54–0.93) and of all-cause death (HR 0.57; 95% CI 0.41–0.80).

What are the clinical implications?

- The totality of randomized clinical trial data suggests that in-hospital initiation of SGLT2i during HF hospitalization may reduce the high risk of clinical events, including all-cause mortality, in the early post-discharge period.

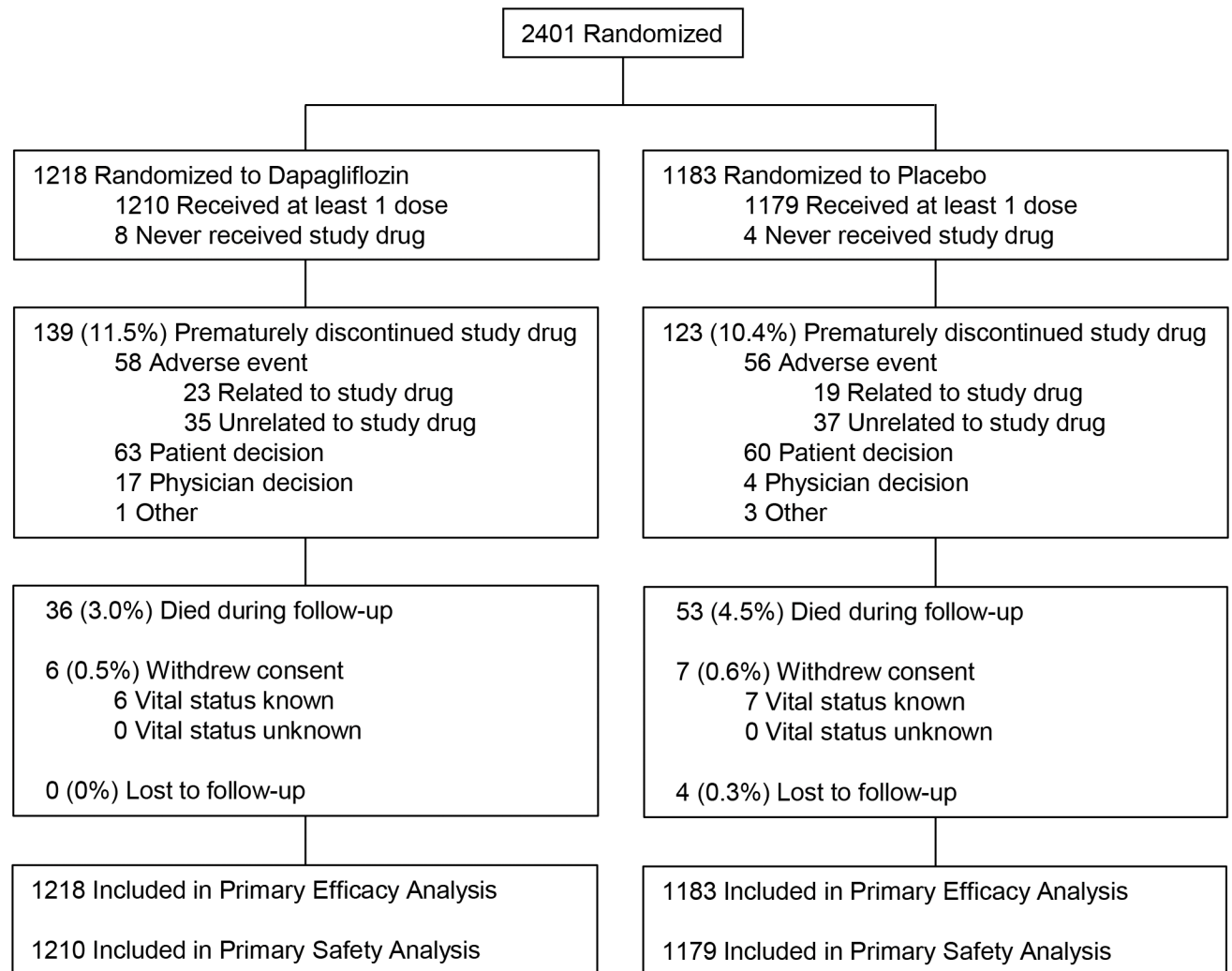
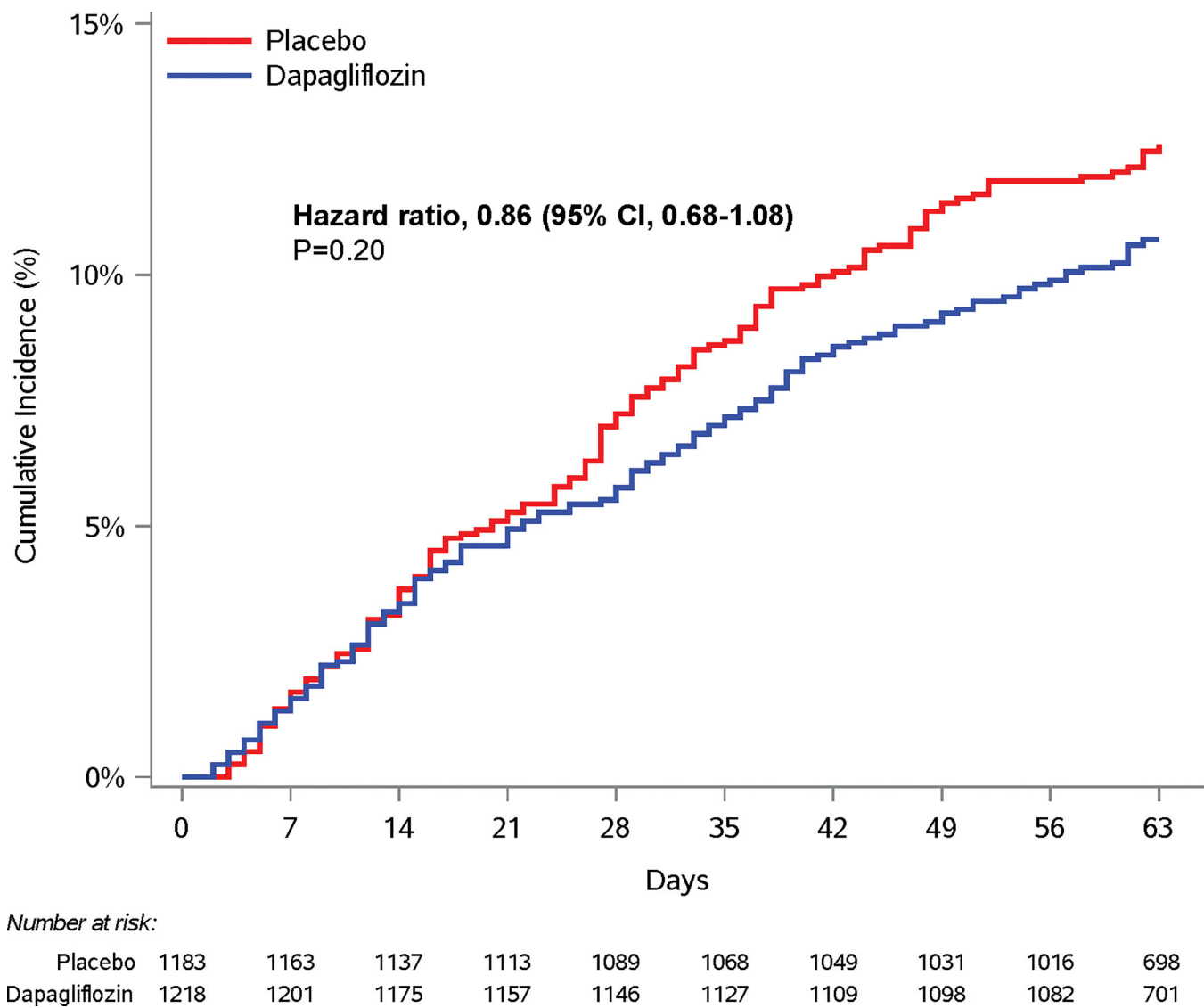


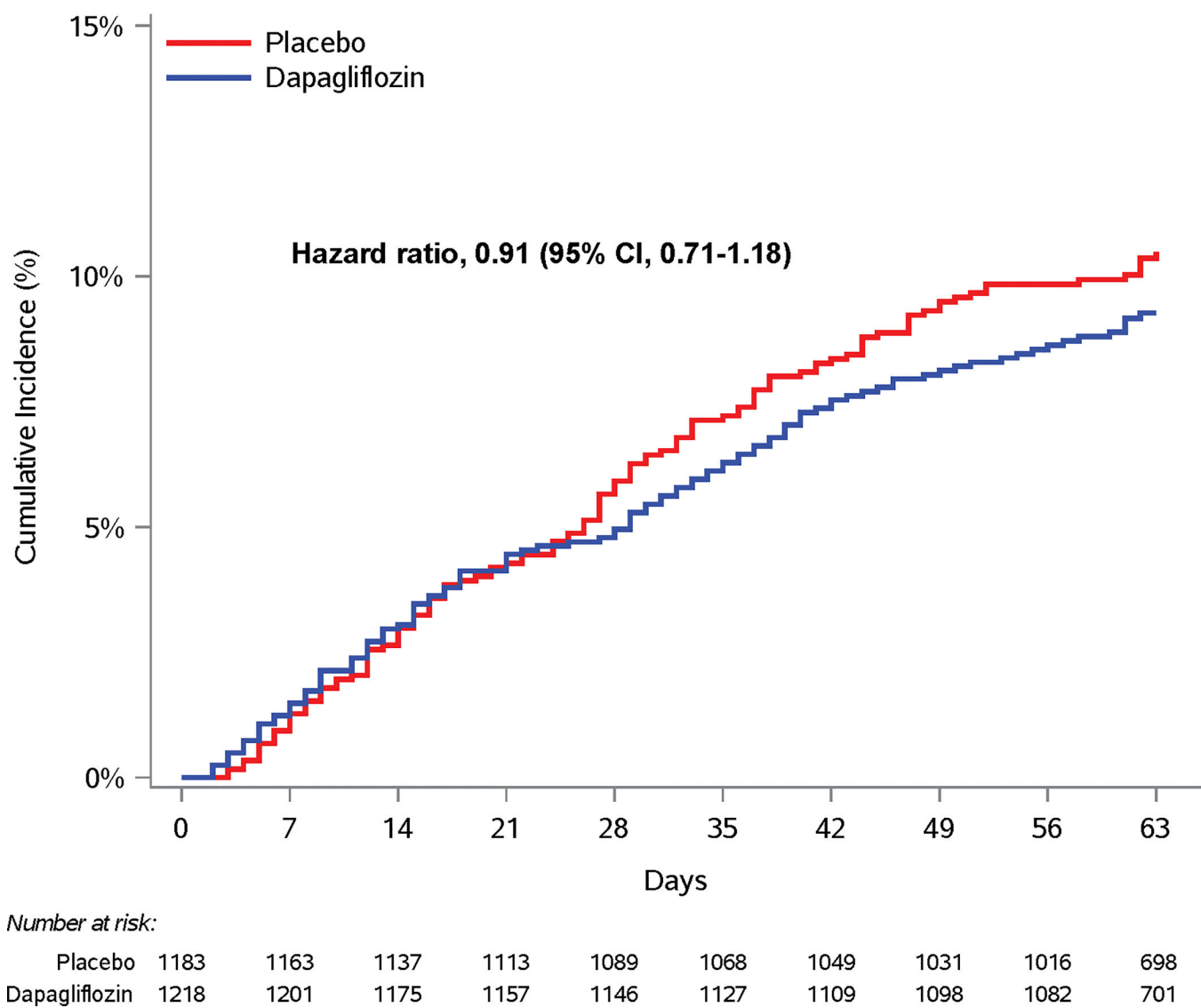
Figure 1. CONSORT diagram.

Two patients were randomized in error and excluded from all analyses. Adverse event relatedness to study drug was based on blinded investigator assessment and includes events related, probably related, or possibly related to study drug.

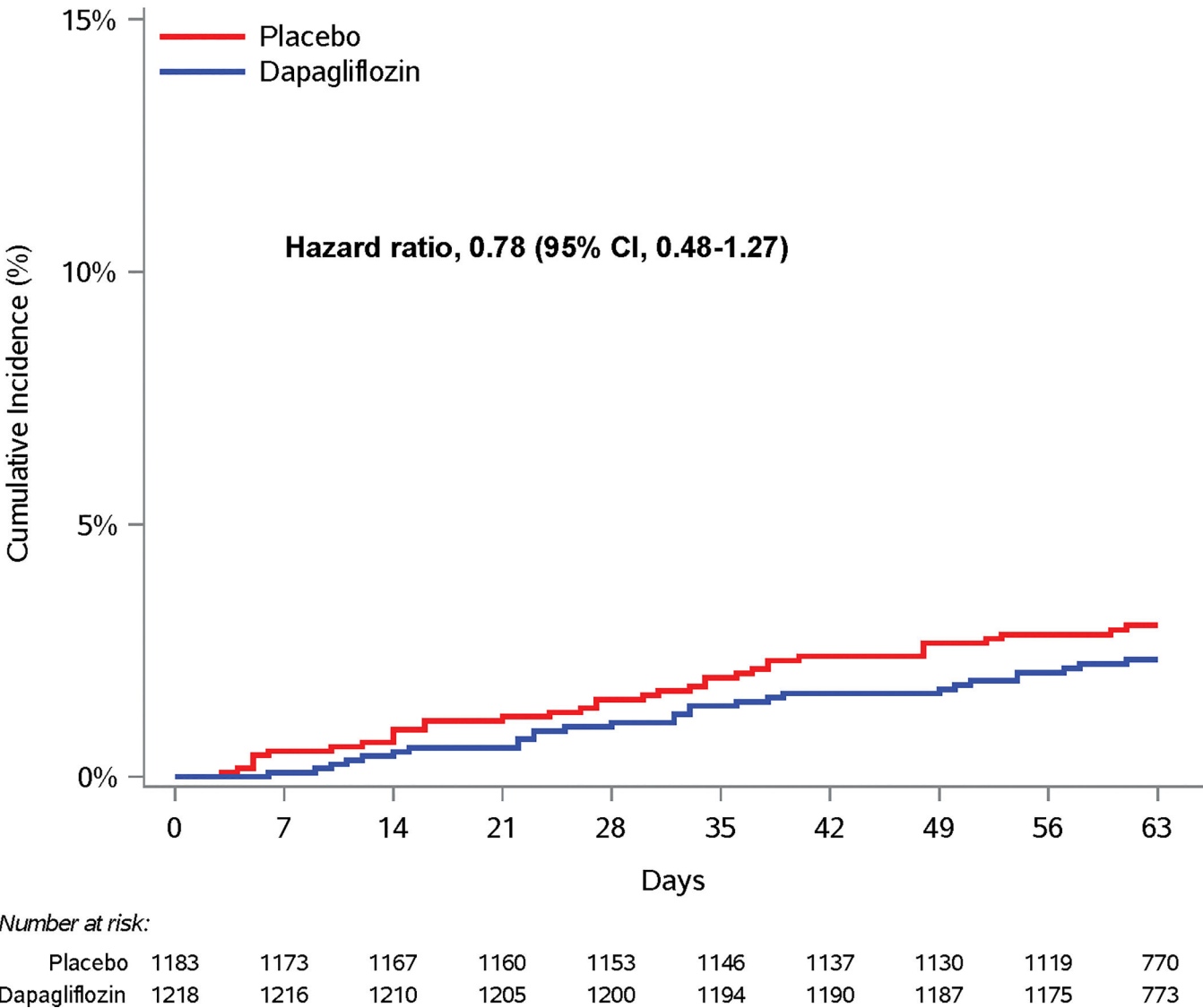
(A) Primary Outcome



(B) Worsening Heart Failure Event



(C) Cardiovascular Death



(D) All-Cause Mortality

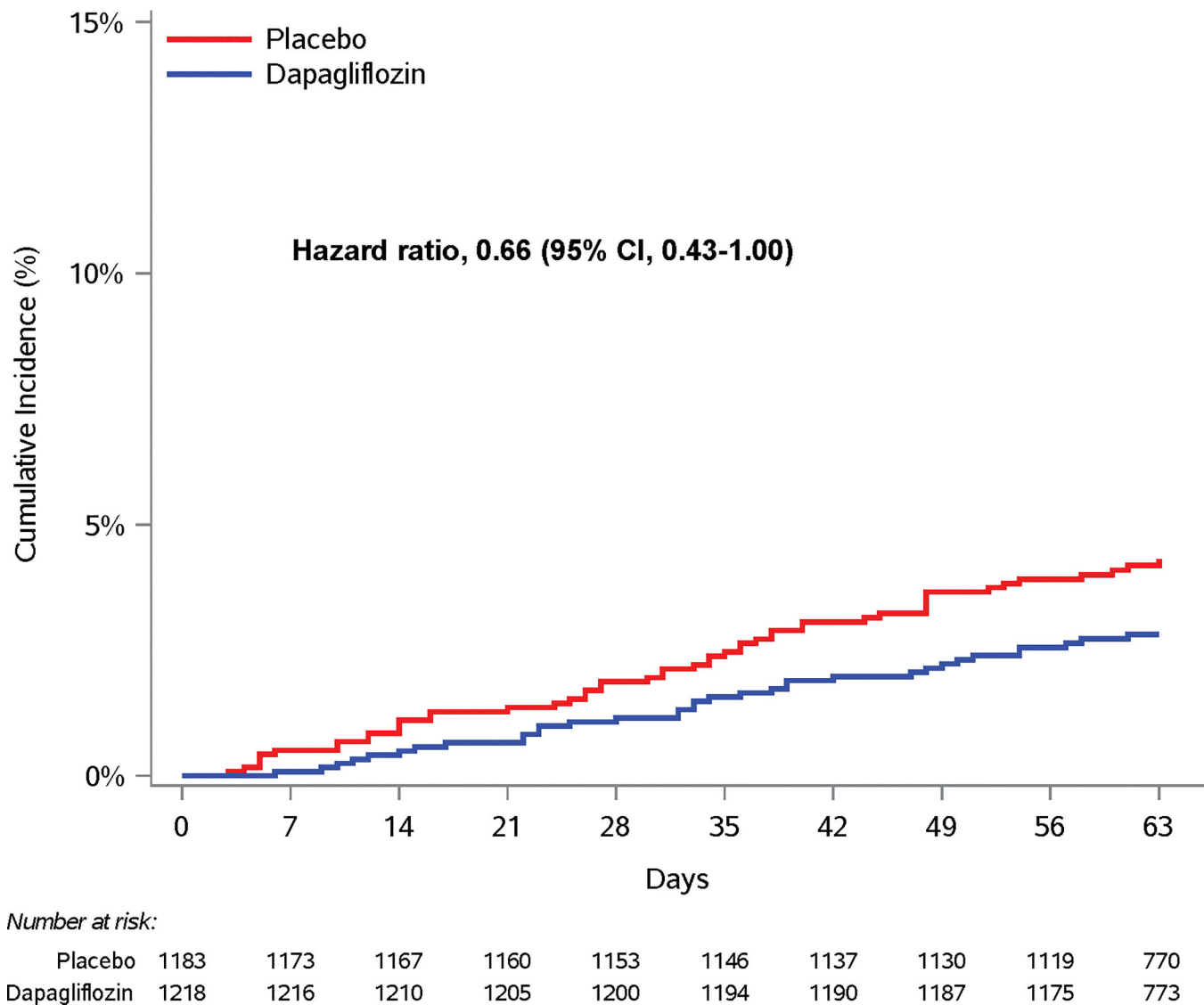


Figure 2. Cumulative incidence of cardiovascular outcomes.
The primary outcome was a composite of cardiovascular death or worsening heart failure. The cumulative incidences of the primary outcome (Panel A), worsening heart failure event (Panel B), cardiovascular death (Panel C), and all-cause mortality (Panel D) were estimated using the Kaplan-Meier method. Hazard ratios and 95% confidence intervals were estimated using stratified Cox proportional-hazards models, with stratification based on type 2 diabetes status and established history of heart failure. The p-value for the primary outcome was calculated using the stratified log-rank test.

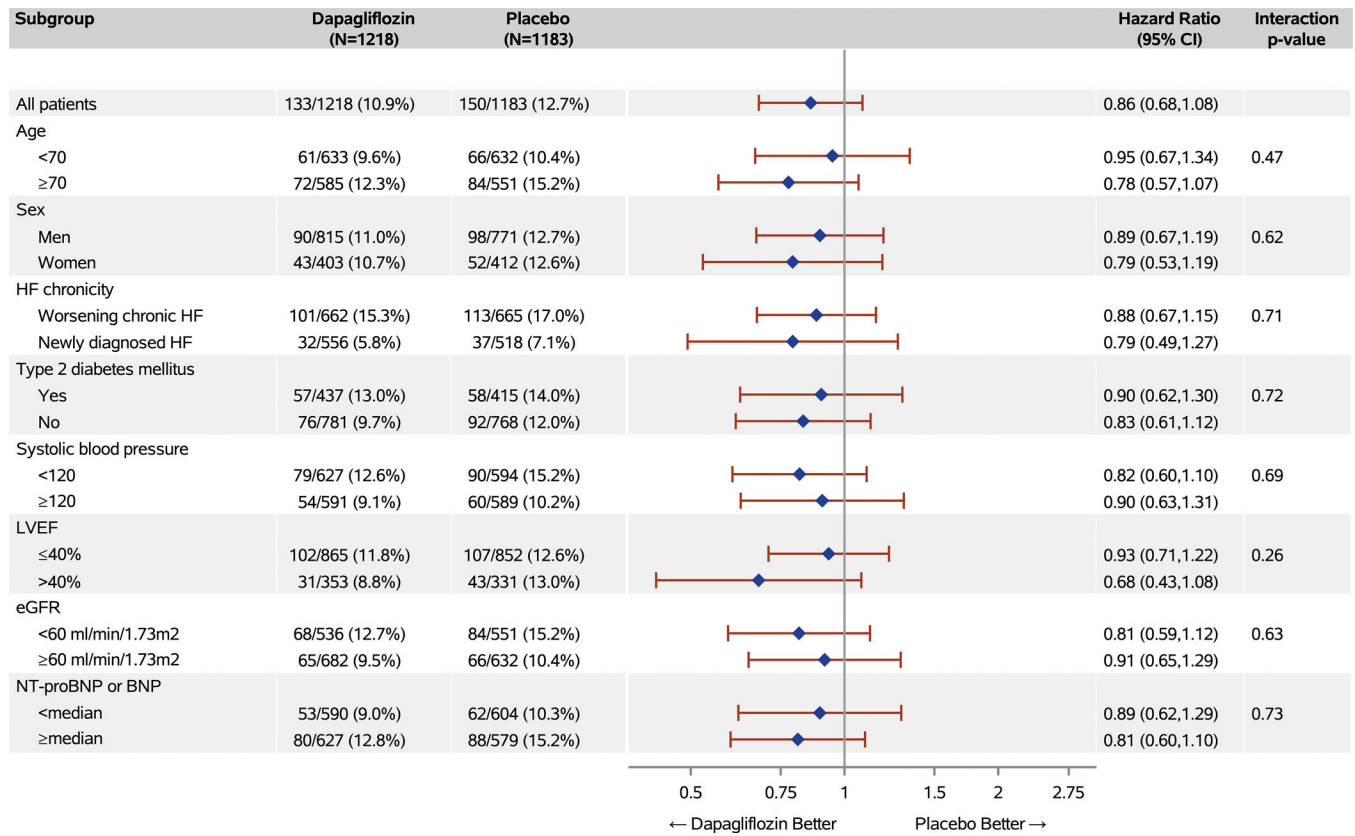
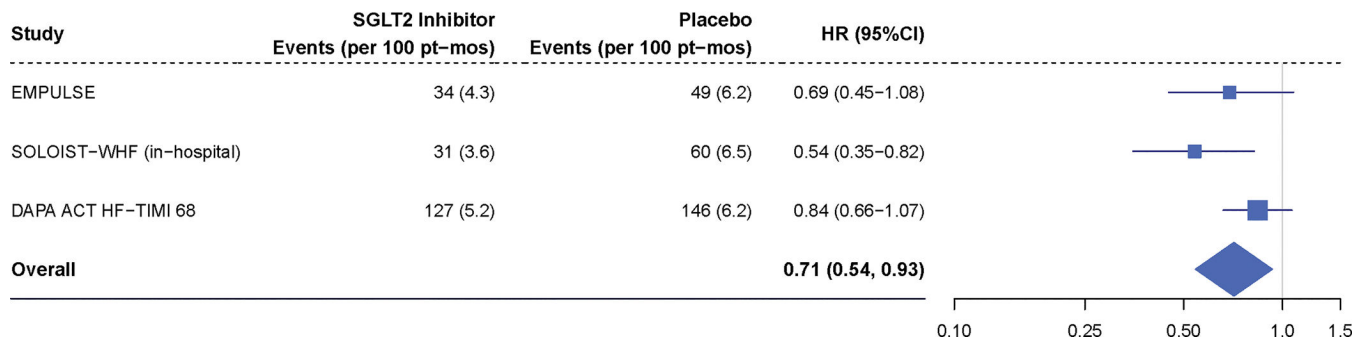
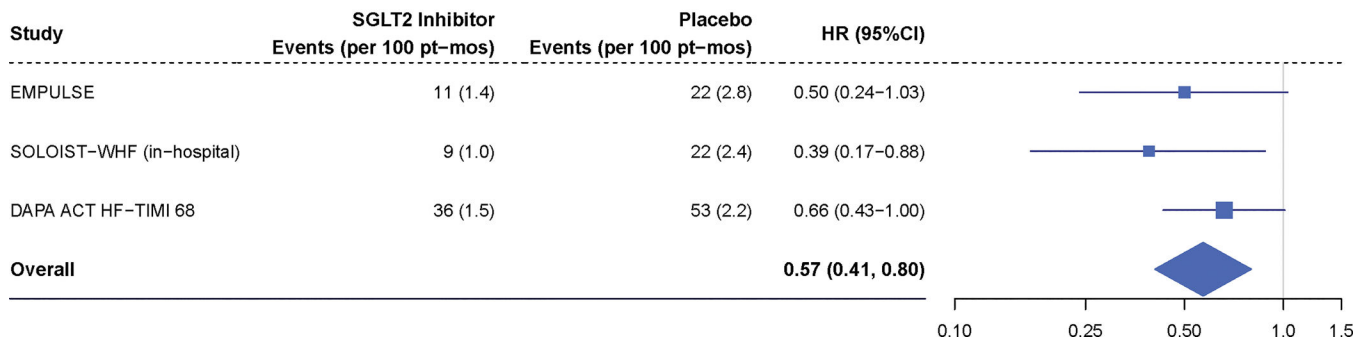


Figure 3. Primary composite outcome in prespecified subgroups.

The forest plot shows the primary composite outcome (cardiovascular death or worsening heart failure) in prespecified subgroups. Variation in treatment effect was assessed with interaction testing using the Cox proportional hazards model.

(A) Cardiovascular Death or Worsening Heart Failure**(B) All-Cause Mortality****Figure 4. Meta-analysis of randomized trials of in-hospital initiation of SGLT2i.**

Pooled estimates of the treatment effect of SGLT2i on the composite of cardiovascular death or worsening heart failure (Panel A) and on all-cause mortality (Panel B), relative to placebo, were generated using random effects models with inverse-variance weighting. There was no between-trial heterogeneity of treatment effect for either the composite outcome ($Q=3.26$; $p=0.20$) or all-cause mortality ($Q=1.40$; $p=0.50$). Pt-mos = patient-months.

Table 1.

Baseline characteristics of study participants.

	Dapagliflozin (N=1218)	Placebo (N=1183)
Age (years)	69 (58–77)	68 (58–76)
Female sex	403 (33.1%)	412 (34.8%)
Body-mass index (kg/m ²)	29.0 (24.9–34.7)	29.5 (25.4–35.4)
Race		
Asian	26 (2.1%)	18 (1.5%)
Black or African American	224 (18.4%)	224 (18.9%)
White	957 (78.6%)	927 (78.4%)
Other	11 (0.9%)	14 (1.2%)
Region		
North America	893 (73.3%)	869 (73.5%)
Europe	325 (26.7%)	314 (26.5%)
Heart failure chronicity		
Newly diagnosed heart failure	556 (45.6%)	518 (43.8%)
Worsening chronic heart failure	662 (54.4%)	665 (56.2%)
Prior heart failure hospitalization	374/662 (56.5%)	368/665 (55.3%)
NYHA class 30 days prior to admission (<i>worsening chronic heart failure patients only</i>)		
I	33/655 (5.0%)	31/658 (4.7%)
II	199/655 (30.4%)	221/658 (33.6%)
III	369/655 (56.3%)	350/658 (53.2%)
IV	54/655 (8.2%)	56/658 (8.5%)
Left ventricular ejection fraction (%)	30.0 (20.0–45.0)	30.0 (21.0–45.0)
LVEF ≤40%	865 (71.0%)	852 (72.0%)
Principal cause of heart failure		
Ischemic	323 (26.5%)	296 (25.0%)
Non-ischemic	727 (59.7%)	716 (60.5%)
Unknown	168 (13.8%)	171 (14.5%)
Type 2 diabetes mellitus	437 (35.9%)	415 (35.1%)
Prior myocardial infarction	243/1196 (20.3%)	222/1167 (19.0%)
History of atrial fibrillation	542/1207 (44.9%)	530/1174 (45.1%)
History of hypertension	943 (77.4%)	935 (79.0%)
KCCQ-12 total symptom score (<i>at randomization</i>)	33.3 (14.6–54.2); N=1215	33.3 (18.8–54.2); N=1180
Natriuretic peptides (<i>qualifying</i>)		
NT-proBNP (pg/ml)	5016 (2867–8766); N=803	4696 (2760–8797); N=779
BNP (pg/ml)	1139 (743–1823); N=421	1092 (701–1889); N=409
Systolic blood pressure (mmHg)	119 (108–132)	119 (108–133)

	Dapagliflozin (N=1218)	Placebo (N=1183)
Heart rate (bpm)	79 (68–89)	77 (68–88)
Estimated GFR (ml/min/1.73m ²)	63.5 (48.5–83.0)	62.5 (47.2–80.7)
<60 ml/min/1.73m ²	536 (44.0%)	551 (46.6%)
Serum potassium (mmol/L)	4.1 (3.8–4.4); N=1218	4.0 (3.7–4.4); N=1181
Medication use (at randomization)		
Beta-blocker	992 (81.4%)	989 (83.6%)
Renin-angiotensin system inhibitor	859 (70.5%)	823 (69.6%)
ACE inhibitor or ARB	539 (44.3%)	533 (45.1%)
ARNI	339 (27.8%)	309 (26.1%)
Mineralocorticoid receptor antagonist	602 (49.4%)	565 (47.8%)
Digoxin	68 (5.6%)	64 (5.4%)
Loop diuretic	1041 (85.5%)	998 (84.4%)

Continuous variables are reported as medians with interquartile ranges, and categorical variables as counts with percentages. Denominators less than the full number of patients in each arm are indicated. ACE = angiotensin converting enzyme; ARB = angiotensin receptor blocker; ARNI = angiotensin receptor-neprilysin inhibitor; BNP = B-type-natriuretic peptide; bpm = beats per minute; GFR = glomerular filtration rate; KCCQ = Kansas City Cardiomyopathy Questionnaire; kg/m² = kilograms per meter-squared; LVEF = left ventricular ejection fraction; ml/min/1.73m² = milliliters per minute per 1.73 * meter-squared; mmHg = millimeters of mercury; mmol/L = millimoles per liter; NT-proBNP = N-terminal pro-B-type-natriuretic peptide; NYHA = New York Heart Association; pg/ml = picograms per milliliter.

Table 2.

Primary and secondary outcomes.

	Dapagliflozin (N=1218)	Placebo (N=1183)	Hazard Ratio (95% CI)
Primary outcome			
Primary composite outcome	133 (10.9%)	150 (12.7%)	0.86 (0.68, 1.08) *
Cardiovascular death	30 (2.5%)	37 (3.1%)	0.78 (0.48, 1.27)
Worsening heart failure	115 (9.4%)	122 (10.3%)	0.91 (0.71, 1.18)
Worsening heart failure during index admission	9 (0.7%)	6 (0.5%)	1.49 (0.53, 4.19)
Re-hospitalization for heart failure	90 (7.4%)	102 (8.6%)	0.85 (0.64, 1.13)
Urgent heart failure visit	21 (1.7%)	23 (1.9%)	0.90 (0.50, 1.62)
Secondary outcomes			
Cardiovascular death, rehospitalization for heart failure, or urgent heart failure visit	127 (10.4%)	146 (12.3%)	0.84 (0.66, 1.07)
Cardiovascular death or rehospitalization for heart failure	110 (9.0%)	133 (11.2%)	0.79 (0.62, 1.02)
Rehospitalization for heart failure or urgent heart failure visit	107 (8.8%)	116 (9.8%)	0.89 (0.69, 1.16)
All-cause death	36 (3.0%)	53 (4.5%)	0.66 (0.43, 1.00)
Hierarchical composite (% wins)	172,221 (42.3%)	157,615 (38.7%)	Win Ratio (95% CI) † 1.08 (0.98, 1.20)
Time to cardiovascular death (% wins)	11,009 (2.7%)	8,440 (2.1%)	
Worsening heart failure events (% wins)	35,061 (8.6%)	31,247 (7.7%)	
Time to first worsening heart failure event (% wins)	1272 (0.3%)	1390 (0.3%)	
Change from baseline in KCCQ-12 TSS (% wins)	124,879 (30.7%)	116,538 (28.6%)	

* p=0.20 for the comparison of dapagliflozin versus placebo (calculated using the log-rank test)

† The hierarchical composite outcome was evaluated using the unmatched stratified win ratio. A win ratio greater than 1 favors dapagliflozin arm.

Table 3.

Safety outcomes.

Safety Endpoint	Dapagliflozin (N=1210)	Placebo (N=1179)
SAE related or probably related to study drug	2 (0.2%)	1 (0.1%)
SAE related, probably related, or possibly related to study drug	24 (2.0%)	21 (1.8%)
Adverse event leading to study drug discontinuation	58 (4.8%)	56 (4.7%)
Symptomatic hypotension [*]	43 (3.6%)	26 (2.2%)
Worsening kidney function [†]	71 (5.9%)	55 (4.7%)
Major hypoglycemia [‡]	3 (0.2%)	3 (0.3%)
Diabetic ketoacidosis [§]	0 (0.0%)	0 (0.0%)

^{*} Symptomatic hypotension included events leading to hospitalization or study drug discontinuation

[†] Worsening kidney function was defined as resulting in at least a doubling of serum creatinine, hospitalization, study drug discontinuation, dialysis, or renal death.

[‡] Major hypoglycemia was defined as resulting in severe impairment in consciousness or behavior, or requiring emergency external assistance

[§] Any clinically apparent diabetic ketoacidosis